Enterprise Cloud Adoption

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Very informative and entertaining!

- A guy who works for Greg

I'll look at it if I have time!

- The management team

What is Cloud?

1) On-demand self-provisioning
   a) It's the users, stupid!

2) Locational independence
   a) Dude, where's my server?
   b) Who cares?

3) Perception of infinite capacity
   a) Actual capacity will be less than infinity.

What is not Cloud?

(and how to tell)

- Virtualization
  - Necessary, but not sufficient.

- Signs you're not using a cloud:
  - No API
  - No self-provisioning of cloud resources
  - It takes greater than 5 minutes to provision a server
    - Same goes for deprovisioning
  - You need to install some special tools to use it
  - Married to a specific hardware/platform/vendor
    - Requires a consulting or heavy services engagement.

- Confused? Does it work like EC2?
  - No: probably not a cloud
  - Yes: probably a cloud.

The Players

Rogue IT

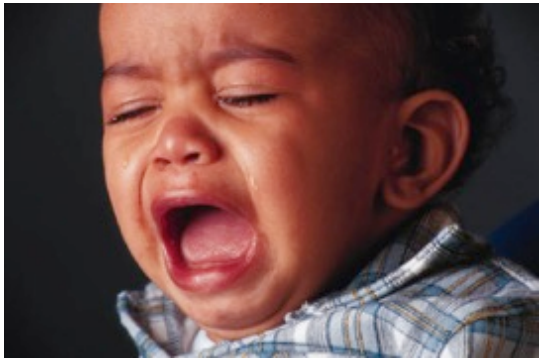
I understand the business, I hear it: to have an open ear, a quick eye, and a nimble hand is necessary for a cut-purse; a good nose is requisite also, to smell out work for the other senses. I see this is the time that the unjust man doth thrive.

Corporate IT

No one knows how old this mighty wizard is, or from whence he came. It is known that, having lived a span far greater than any normal man's, he grew weary of lesser mortals; and so, spurning all human company, the forsook the dwellings of men and went to live in the depths of the Earth.

The Customer

These strange creatures live mostly on the surface of the earth, gathering together in societies of various forms, but occasionally a stray will descend into the depths and commit mayhem among the dungeon residents who, naturally, often resent the intrusion of such beasts. They are capable of using weapons and magic, and it is even rumored that the Wizard is a member of this species.

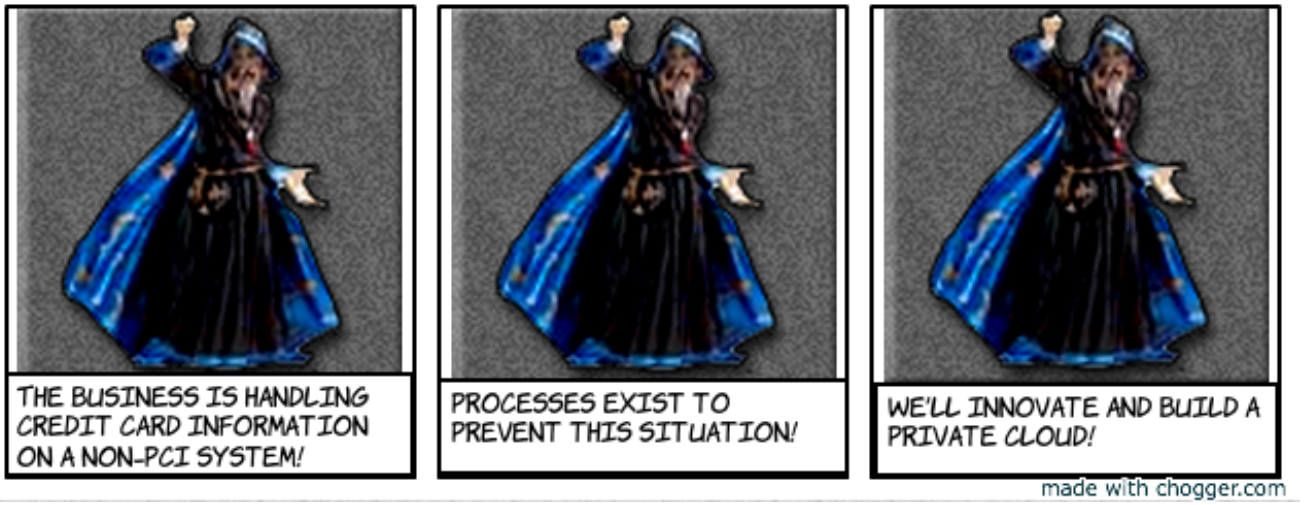


- The Tragedy -

… 9 *minutes* later…

1. Provision a VM from $public_cloud_vendor
2. Use image containing eCommerce software
3. Code deep into the night.
4. ~~Test~~?
5. Push!
6. Production ready!
7. Ship it!
8. Credit card orders roll in.
9. ???
10. Profit?

- The Comedy -



… 9 *months* later…

1. Google for private cloud vendors.
2. Spend millions to develop a shim on top of the virtualized data center.
3. Focus entirely on dev/test use cases.
4. Impose existing IT processes onto the cloud.
5. Advertise your cloud!
6. Write a whitepaper espousing the success!
7. ???
8. Punish the innocent.
9. Praise and honor for the non-participants.

- Rogue IT -

Agility >> Control

- Agility matters, perhaps more than anything.
- Process is a barrier to creativity.
- Seeks to limit risk, but doesn't want to sacrifice value in doing so.
- Is not an expert in risk management.
- Seeks to have her workflow be understood and supported.
- Cares deeply about delivering value to customers by using a narrative of agility and scalability.

- Corporate IT -

Control >> Agility

- Working to control risk while operating at tremendous cost pressure
- If credit card number are stolen:
  - The business will suffer
  - She will lose her job
- Endeavors to make risk management scale using rich, time-tested processes.
- Seeks to understand customer stakeholders.
- Cares deeply about delivering value to customers by using a narrative of predictability, reliability, and security.

- They Said, They Heard -

I need cloud!

I have built you a cloud!

Something like AWS!

Virtualized Data Center!

- They Said, They Heard -

Self-provisioning? I can create my own VM?

Once I get a VM, I will cherish it forever!

Or, I can be more agile in my build/ deploy process.

Self-provisioning? Users can create their own VM?

We'll run out of capacity in no time flat!

A curse upon this cloud project!

- They Said, They Heard -

This VM is months old. I'd like to start fresh, but it takes so long to get a new server.

The cloud is supposed to be about agility!

Cloud computing is a failure.

These VM are months old, and no one is using them.

The cloud is supposed to be about agility.

Private cloud is a failure.

- Assurance and Admiration -

Well, maybe.

It’s the users, stupid!

Users feel empowered, managers feel informed.

On-demand, self-provisioning.

Approval processes and paperwork (virtual or otherwise) kills cloud.

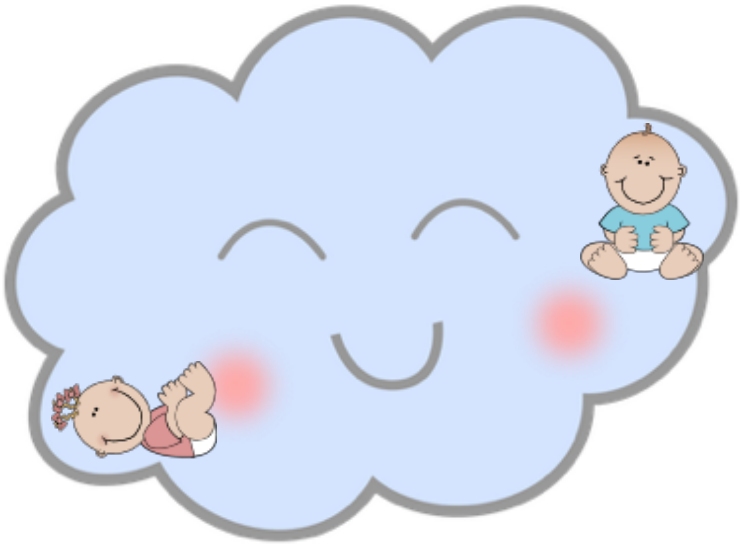
- The End -
Do you believe in the users?

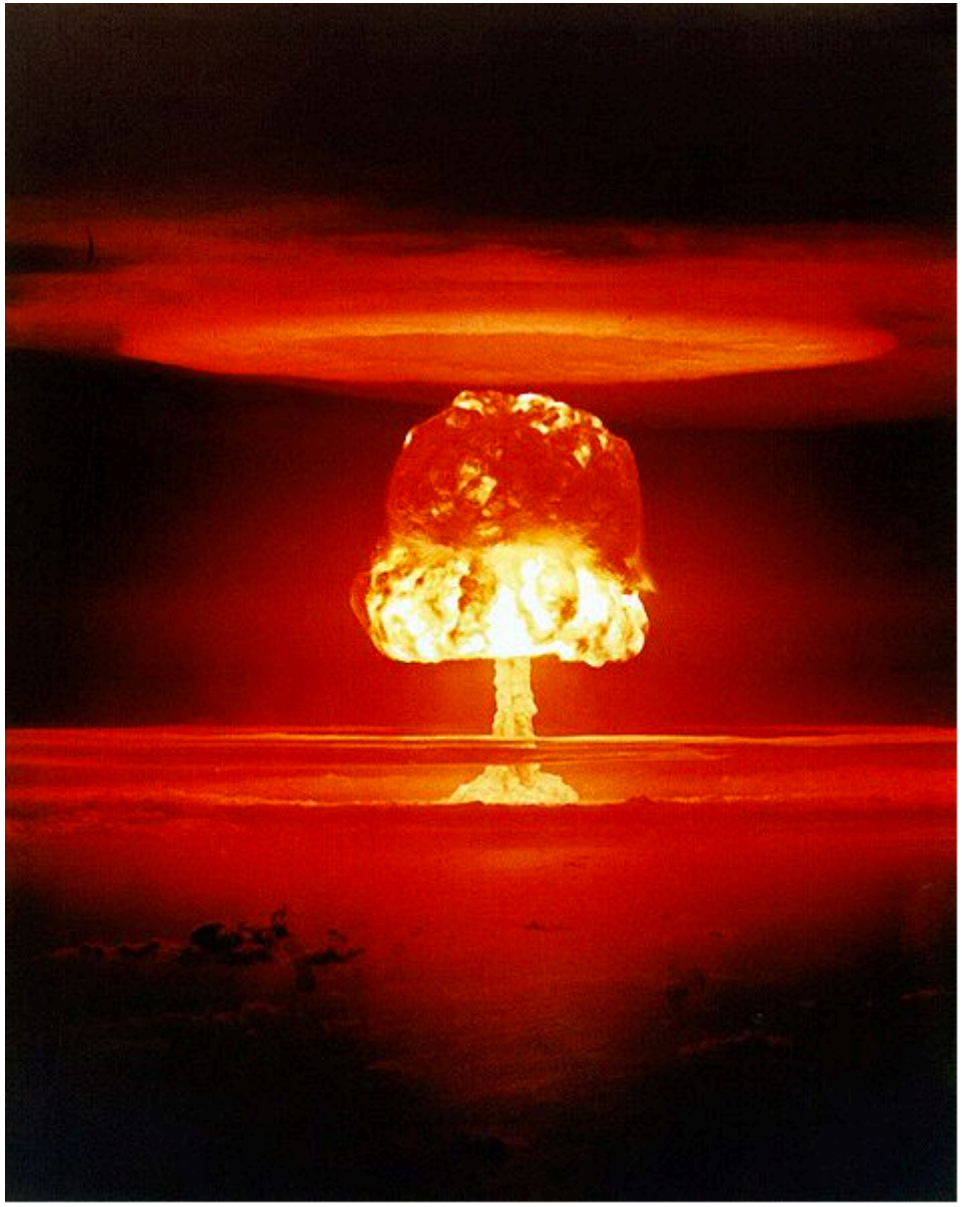
Okay, don’t.

(I wouldn’t) Let’s see some proof.

What to Expect WhenYou're Expecting Cloud

Let's Panic About Cloud!





Executive Summary

The Panic Part

Mature, cloudy enterprise will take time.

2 year path to fully matured adoption.

Mature, cloudy enterprise will take time.

The transformation process should be disruptive, positively.

Mature, cloudy enterprise will take time.

End users should feel empowered by the transformation, managers should feel informed and in control.

Mature, cloudy enterprise will take time.

Phased Approach

1. Breaking Cloud (0-6 months)

The desired outcome of this phase is the the formation of a staged, reasonable, risk-aware transformation project.

2. Refining Process (6-18 months)

The desired outcome of this phase is the on-boarding of critical user groups/processes, limited in scope, but willing to learn and assist the adoption tiger team in refining the strategy.

3. Cloud Norming (18-24 months)

The desired outcome of this phase is that the onboarding process is well-established, scalable, and evidenced by a growing user base.

4. Cloud Scale (24+ months)

The desired outcome of this phase is a fully transformed organization. Users are empowered to self-provision cloud resources, managers have good visibility into usage, and the team considers the last 24 months' effort "worth it."

Cloud Tiger Team

1) Pays down technical debt during cloud adoption.

2) This team should be disruptive, empowered, and responsible. Attributes include:

a) Adventurous, willing to drive adoption of new technologies.

b) Problem solvers. Process oriented.

c) Accountable. Meaning they make data-driven decisions.

d) Visionary. Never lose sight of the desired end state: user enablement.

e) Invested. Can't just say: "The cloud is broken! Call the vendor to fix it!"

3) Identify a cloud adoption team and empower them.

a) Hint: Your vmware experts are probably *not* your cloud adoption experts.

The VMware Paradigm

User: I want to provision a virtual machine.

VMware Admin: Okay, what cluster do you want it on? Do you want to thin or thick provision? Thick? Ok, eager or lazy zeroed? Which data store do you want to use? Which OS image do you need? Ok, I've put in the ticket with the networking team. You should have the VM in 3 weeks…

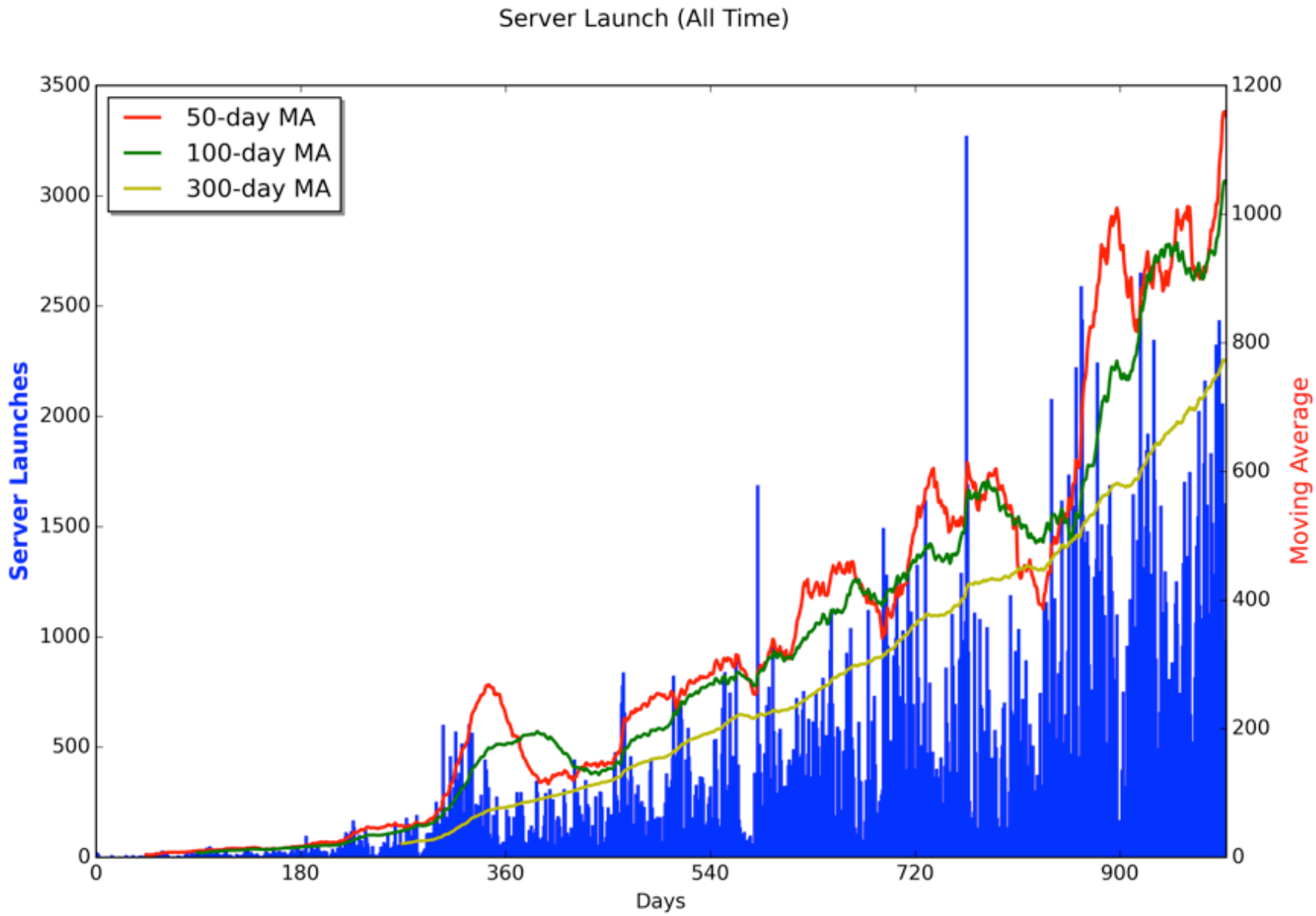
User (eyes glazed): Well... Can I just get a VM?

VMware Admin: That's what I'm trying to give you!!!!

User: Forget it, I'll just go use AWS.

To the user/consumer, this feels nothing like AWS.

Cloud Adoption, by the Numbers



Cloud Adoption, by the Numbers

1. Server Launches

Gives a measure of cloud “utilization”. Not necessarily by the amount of resources consumed (CPU/Memory/Disk) but by how many “customer transactions” of this type have occurred.

2. Server Lifetime

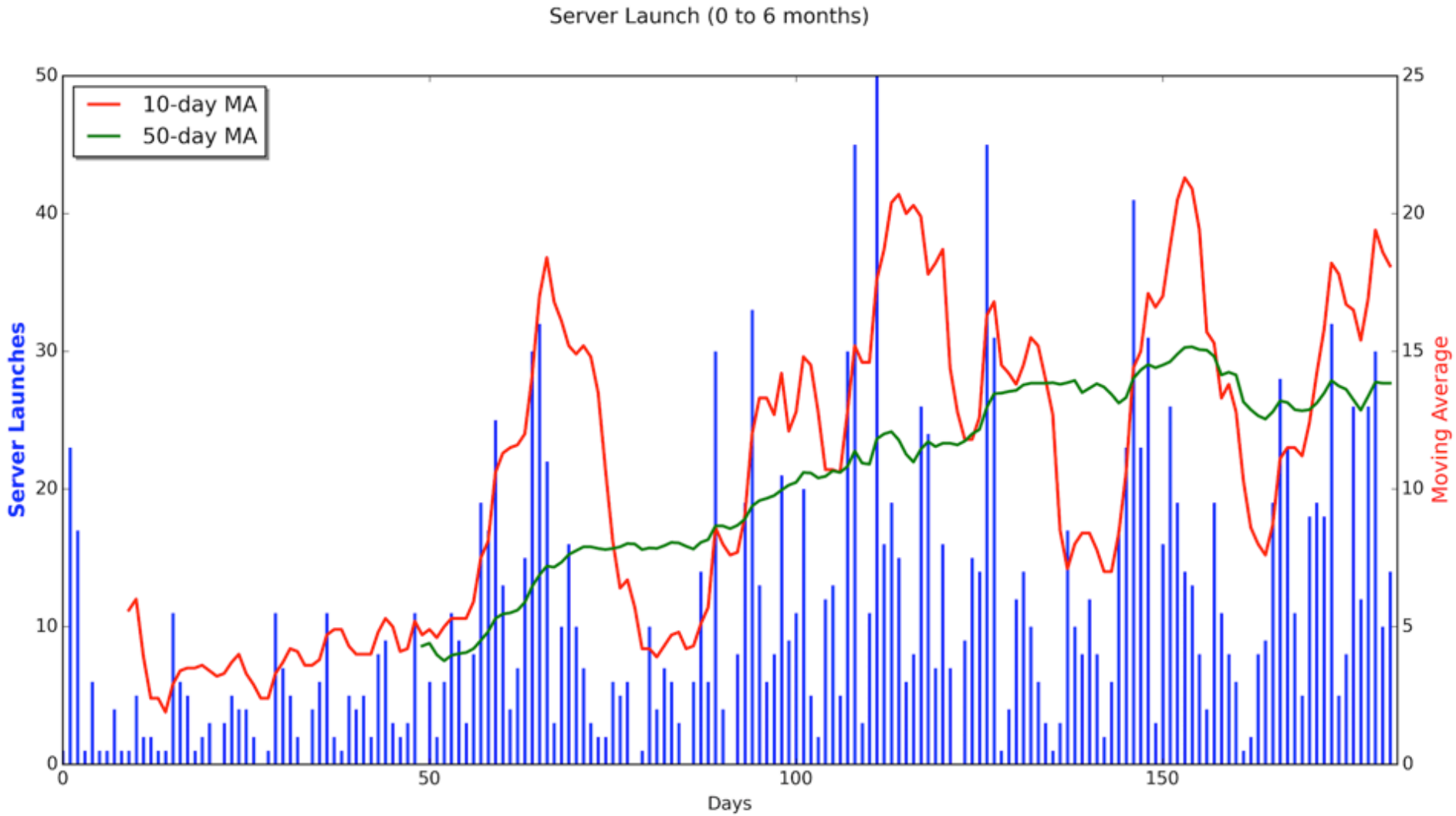
Provides insight into how transient users consider compute and other IaaS resources to be. This metric is most appropriate in software development.

Focus shifts to the *process* used to arrive at a functional compute resource (Chef/Puppet)

3. User Activity

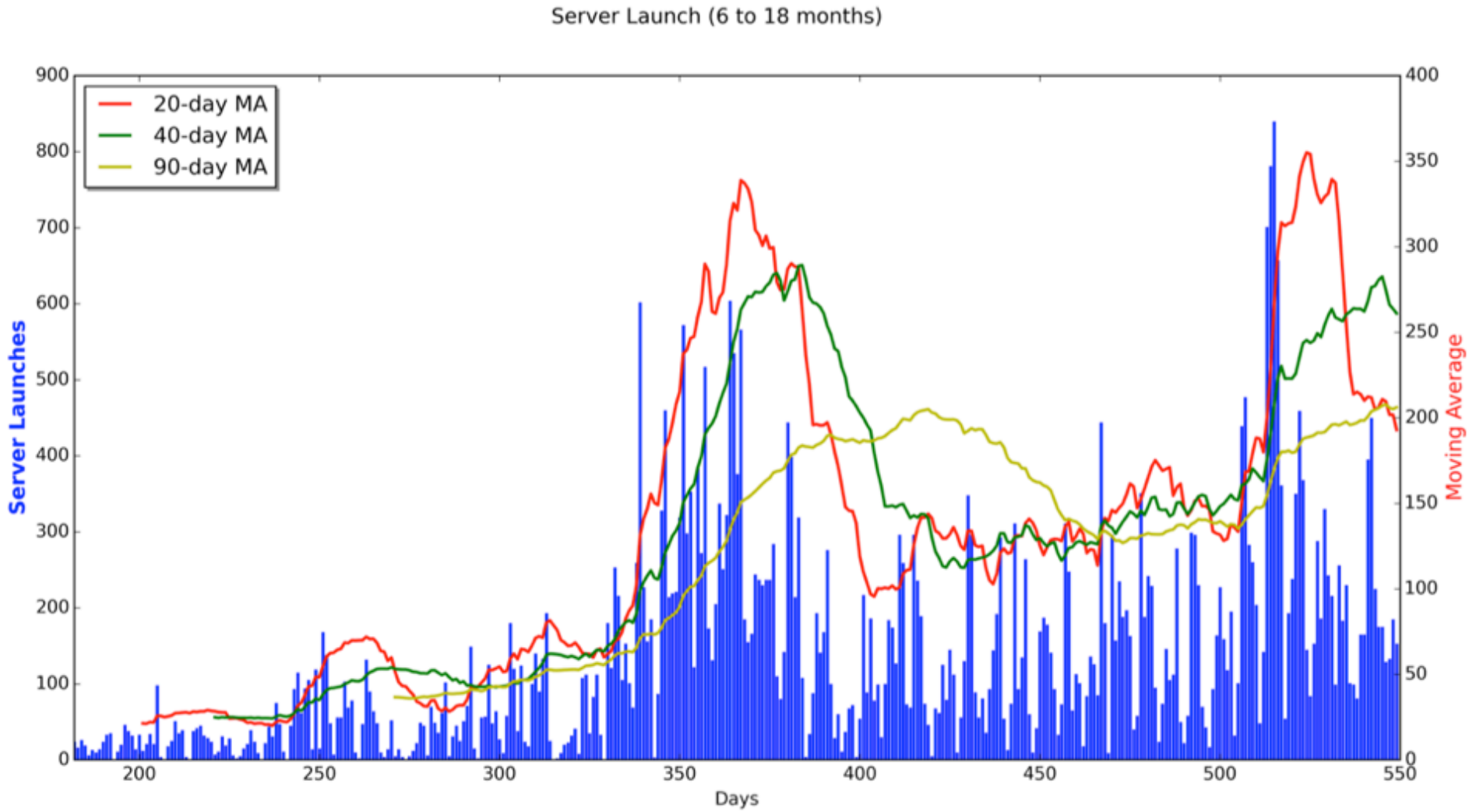
Helps answer the question: How successful was the transformation?

0-6 months – Breaking Cloud



The desired outcome of this phase is the the formation of a staged, reasonable, risk-aware transformation project.

6-18 months – Refining Process



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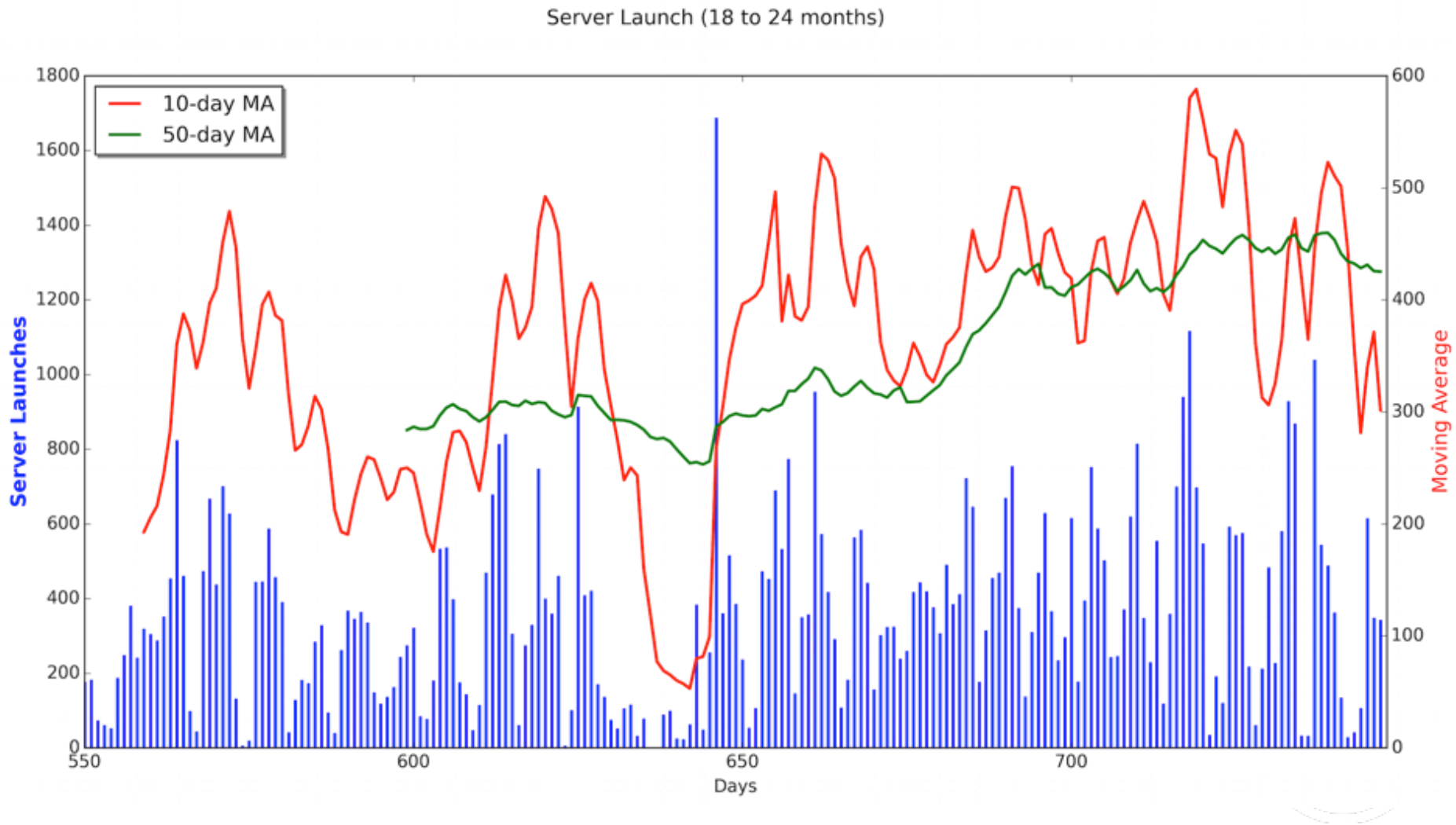
Limited in scope, but willing to learn and assist the tiger team in refining strategy.

18 - 24 months – Cloud Norming



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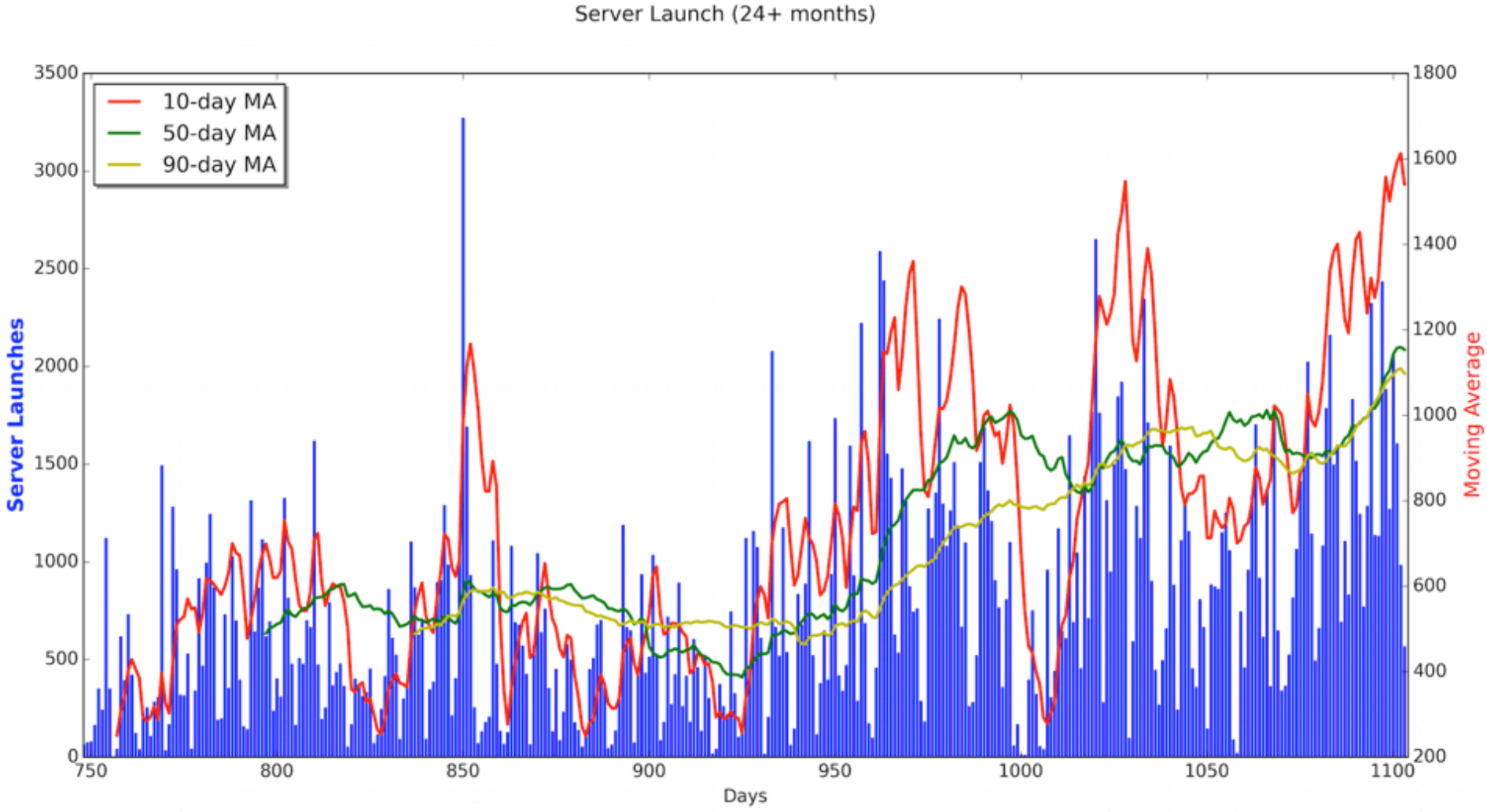
24+ months – Cloud Scale



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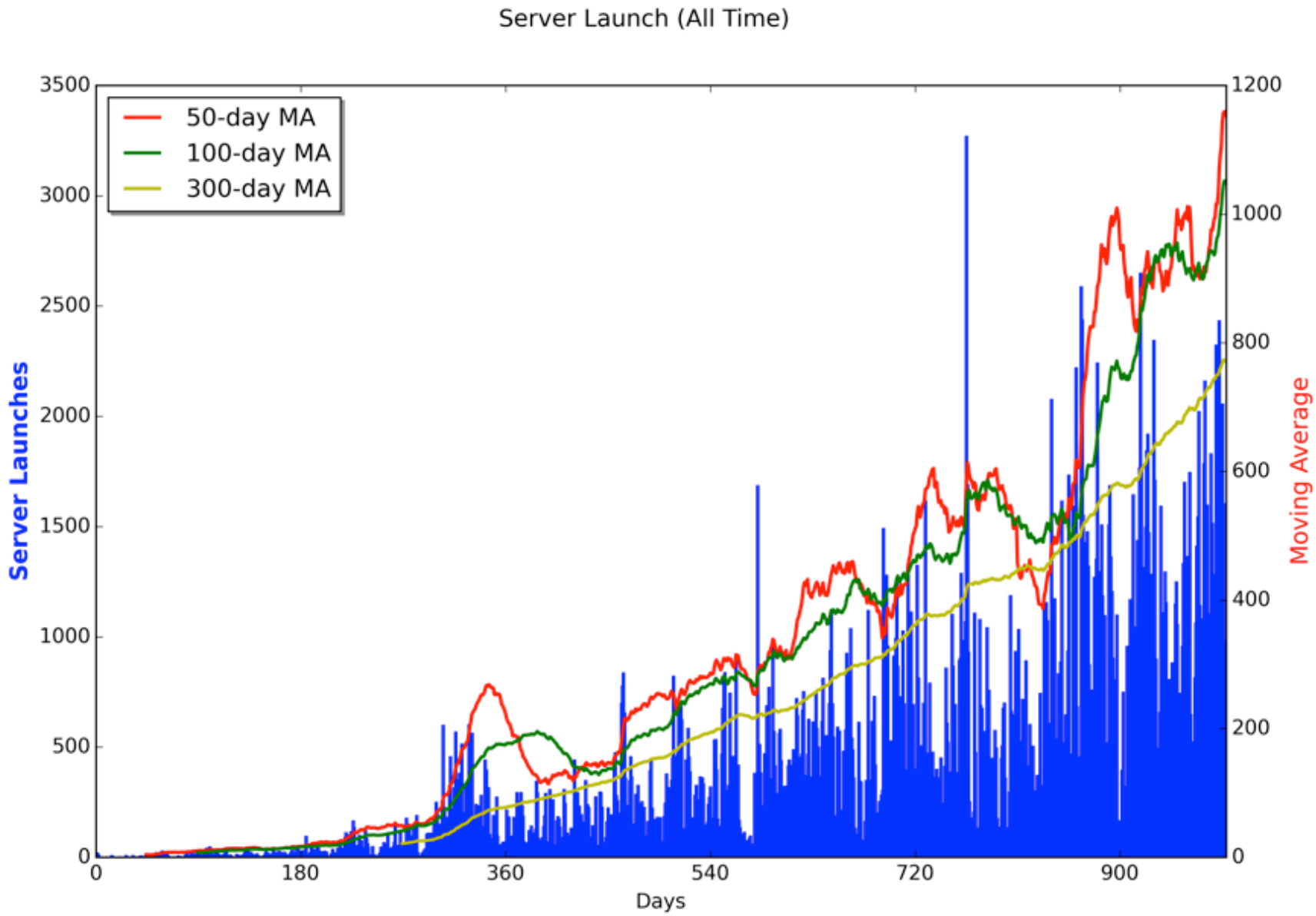
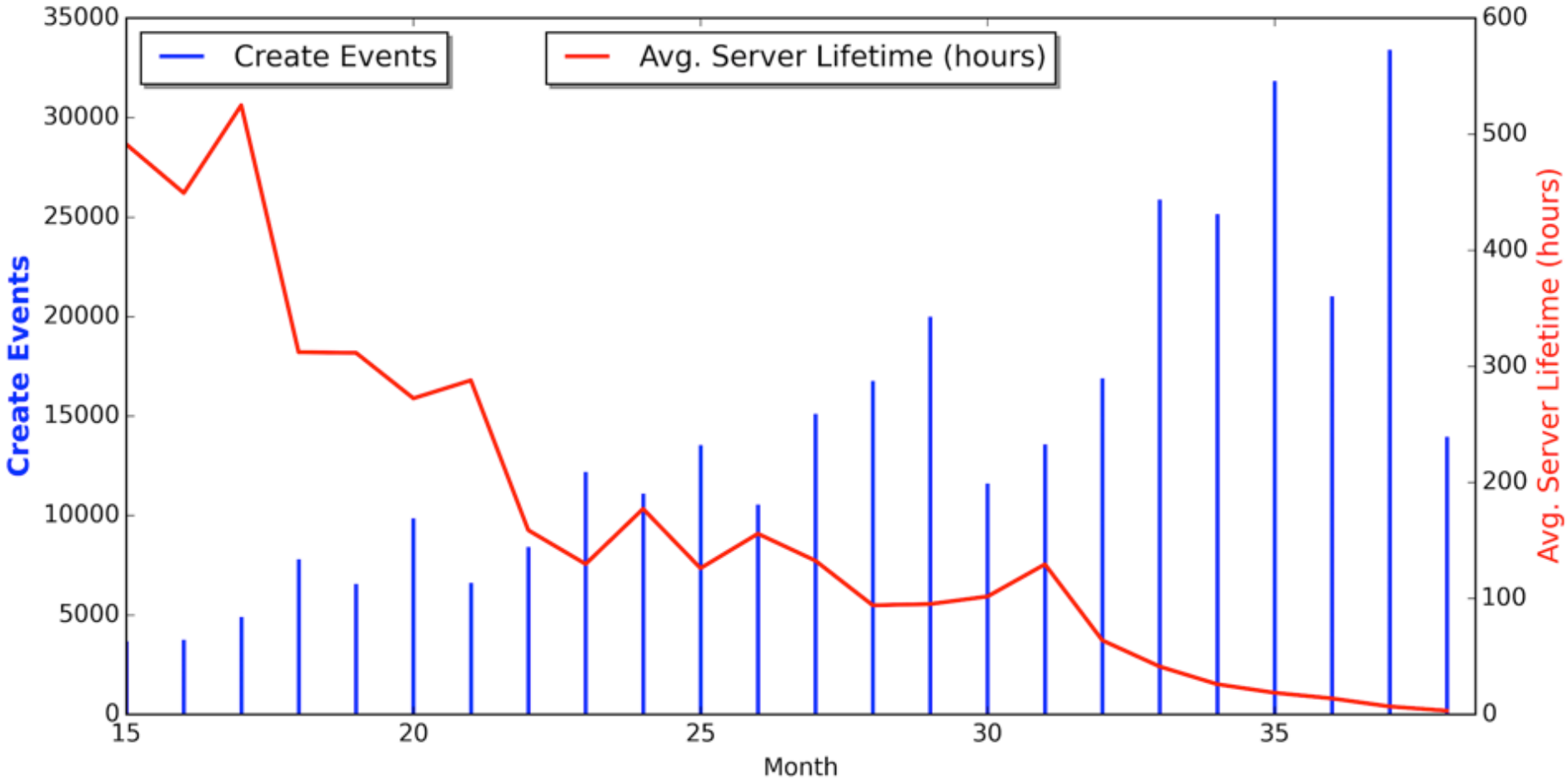
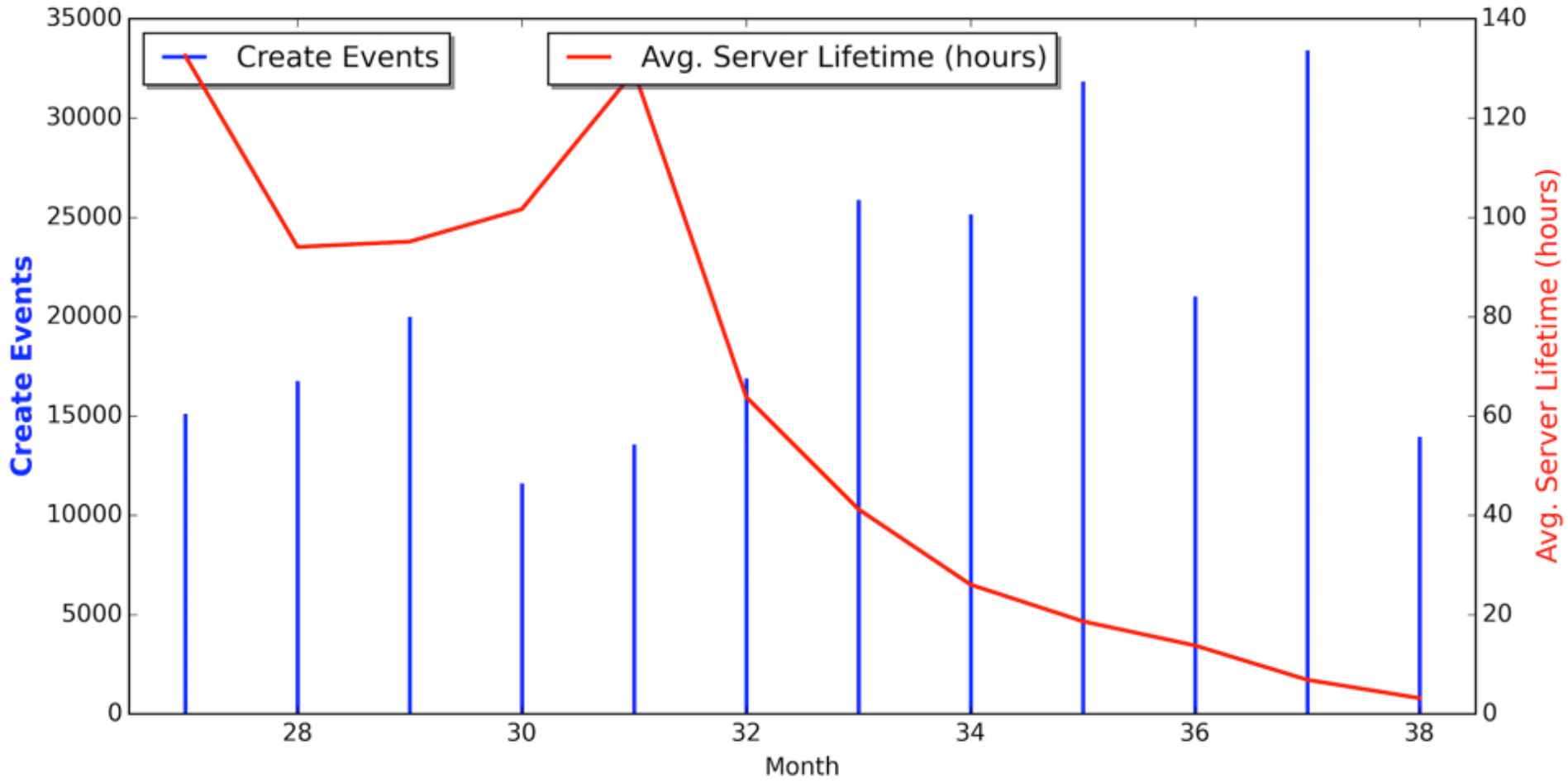
Cloud Adoption, by the Numbers



Server Lifetime



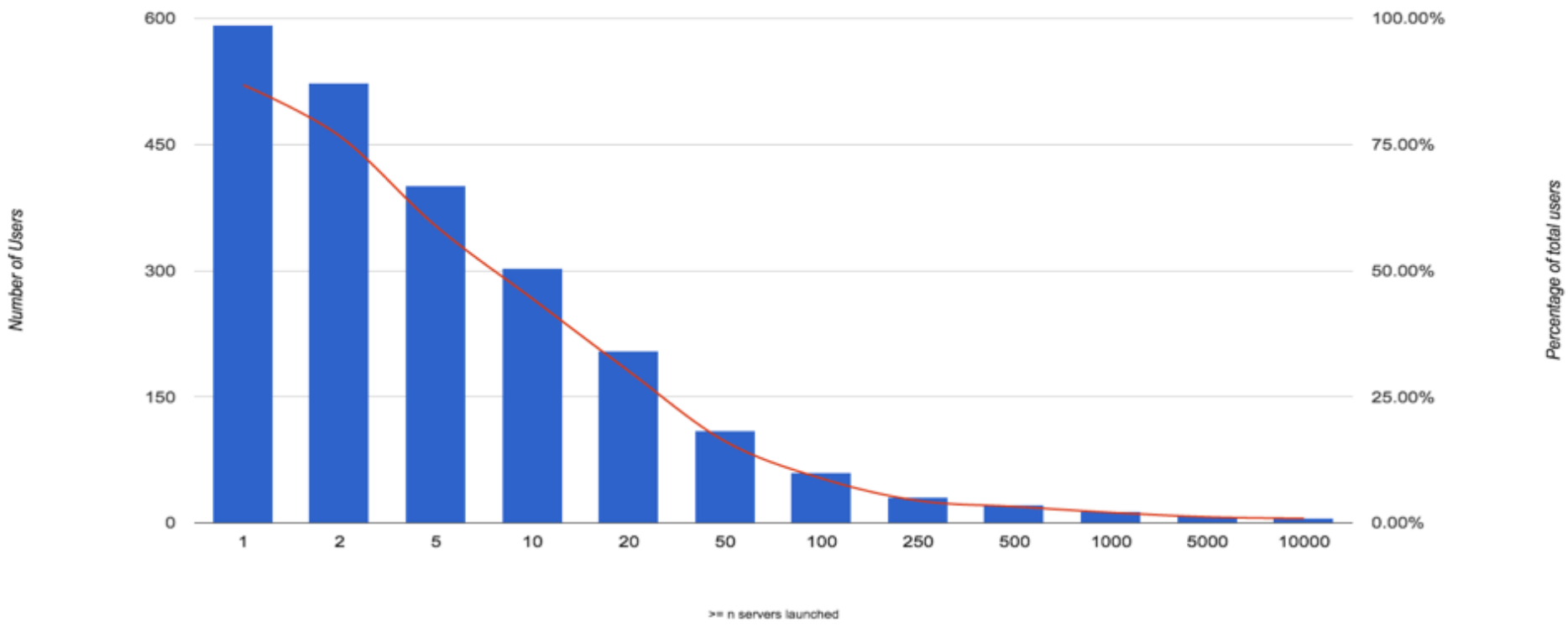
Server Lifetime (last 12 months)



Users

- Total users: 1214 (682 Active)
  Active = Launched or modified at least 1 server.

n	1	2	5	10	20	50	100	250	500	1000	5000	10000
Users	592	523	401	303	205	110	60	30	22	14	8	6
%	86.8	76.69	58.8	44.43	30.06	16.13	8.8	4.4	3.23	2.05	1.17	0.88



Summary

Enterprise cloud adoption is as much about cultural paradigms as it is about technology.

Even when, or perhaps especially when done correctly, the transformation will take time.

User/Customer enablement is the goal.

Metrics will help inform the strategy.

- The End –

For real this time.

Thank you!

Bonus material ahead, time permitting

Users (Accounts)

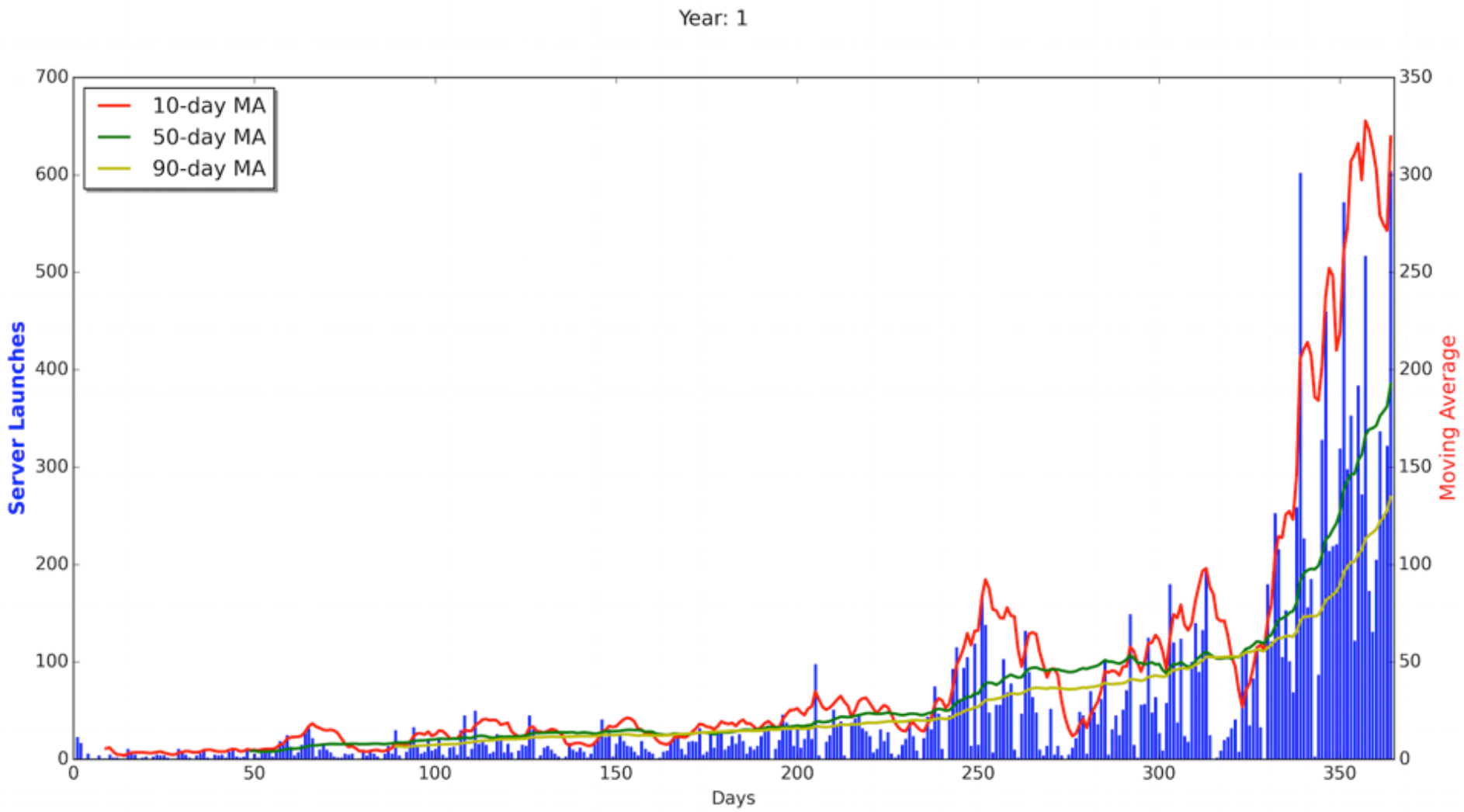
Number of Servers Launched/Running per Account

Launched:

Min	Max	Mean	Median	Std. Deviation
1	86183	1692.30	19.0	8105.12

Running:

Min	Max	Mean	Median	Std. Deviation
0	609	26.28	1	72.54


Year: 1
10-day MA
50-day MA
90-day MA
Server Launches
Moving Average
Days
0
100
200
300
400
500
600
700
0
50
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150
200
250
300
350
0
50
100
150
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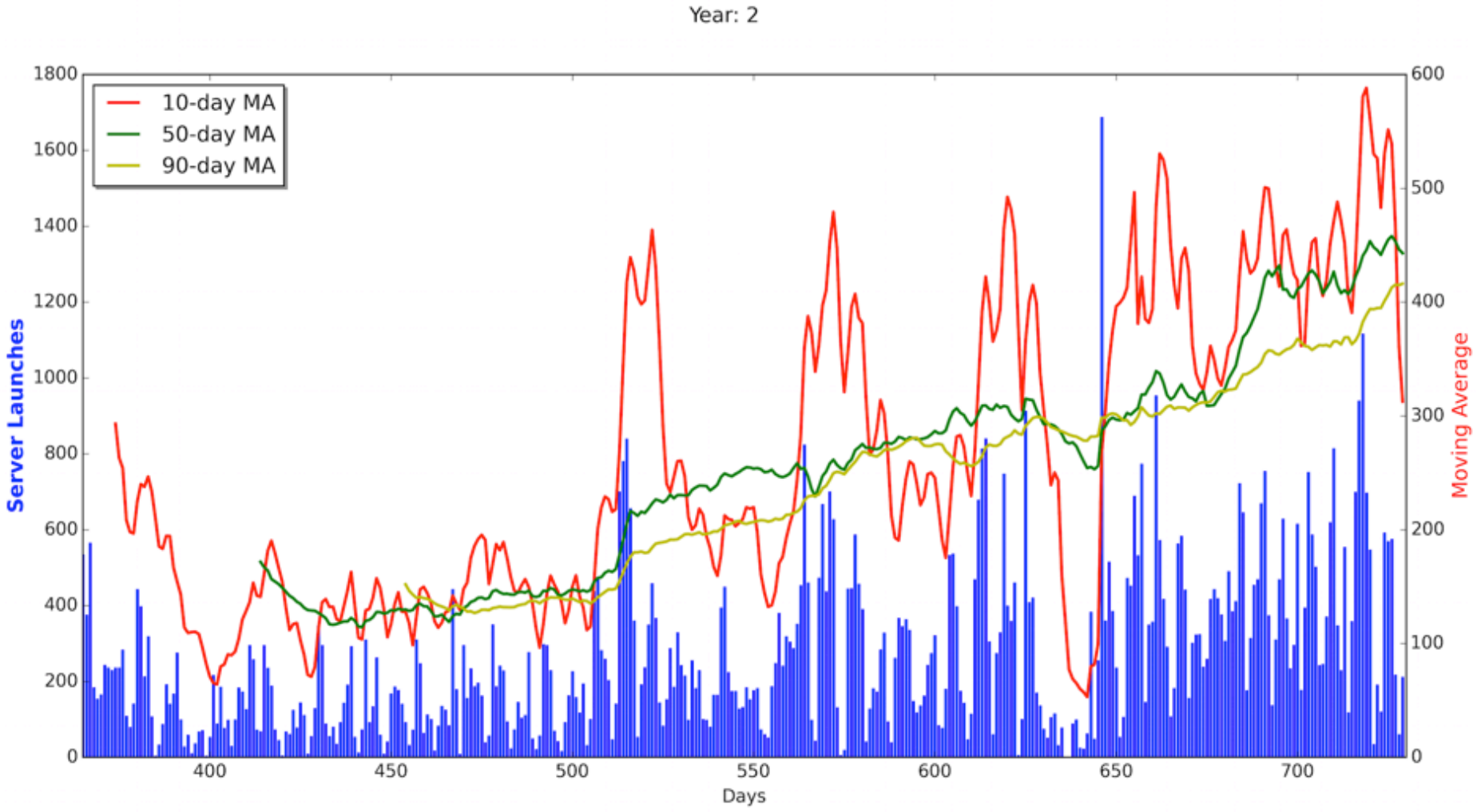
250
300
350


Year: 2
10-day MA
50-day MA
90-day MA
Server Launches
Moving Average
Days
0
200
400
600
800
1000
1200
1400
1600
1800
0
100
200
300
400
500
600
400
450
500
550
600
650
700

Attribution

1) Slides 5, 6, 7, 9, 10, 11
   a) Rogue (Guy Fawkes) http://en.wikipedia.org/wiki/Guy_Fawkes#/media/File:Guy_Fawkes_by_Cruikshank.jpg
   b) Wizard (of Yendor)
      i) http://groups.google.com/group/rec.games.roguelike.nethack/msg/8367e3685d771c42
      ii) http://www.aesthetictech.8m.com/absurd.htm
2) Slide 12
   a) Virgin and the Unicorn, http://en.wikipedia.org/wiki/Unicorn#/media/File:DomenichinounicornPalFarnese.jpg
3) Slide 9,14
   a) Mushroom cloud
      i) http://commons.wikimedia.org/wiki/File:Castle_Romeo.jpg
   b) Cloud Babies
      i) https://openclipart.org/
4) Slides 6, 7
   a) Comic strips
      i) http://chogger.com
5) Slides 6
   a) Guy Fawkes mask
      i) http://en.wikipedia.org/wiki/Guy_Fawkes_mask#/media/File:GuyFawkesMask.jpg, http://creativecommons.org/licenses/by-sa/3.0/
6) Slide 9
   a) AWS Simple Cloud
      i) http://commons.wikimedia.org/wiki/File:AWS_Simple_Icons_AWS_Cloud.svg